

Initial D Arcade Stage Zero v2 BGM AutoMod Guide

Automated music replacement workflow by Metasharp (evolved from PockyWitch one)

Updated for the current Zero v2 script, the completed ID8 AutoMod guide structure, and the 01-26 song mapping spreadsheet.

Purpose. This guide replaces the older mostly-manual D0 music replacement tutorial with the current CMD-based workflow. The script handles backups, extraction, WAV conversion, HCA encoding, ACB/AWB rebuilding, optional UI texture patching, mod-output packaging, and publishing to the selected game folder.

1- Modding Folder Structure

```
./initial_d_arcade_stage_zero_v2_bgm_automod.cmd
./backup/
./mod-output/
./new-musics/
./new-musics-previews/
./new-musics-gfx/
./tmp/new-wav/
./tmp/new-wav-prev/
./tmp/hca/
./tmp/hca-prev/
./tmp/AVEX/
./tmp/ADX_SELECT/
./tools/ffmpeg/
./tools/vgaudio/
./tools/SonicAudioTools/
```

2- What the Script Automates

1. Game folder selection and saving for future runs.
2. Backup and restore for the modified ACB/AWB and PAC files.
3. Extraction of ADX_SELECT and AVEX with SonicAudioTools AcbEditor.
4. Optional automatic WAV generation from new-musics if tmp/new-wav is empty after the manual audio step.
5. Optional manual loudness normalization with Audition, Audacity, or another editor.
6. Conversion of full songs to looped HCA files with VGAudio.
7. Preview generation: supplied previews are used when available; otherwise 50-second previews are generated from tmp/new-wav.
8. Conversion of preview songs to non-looped HCA files.
9. Rebuilding of ADX_SELECT.acb/.awb and AVEX.acb/.awb.
10. Optional publishing of UI texture PAC files.
11. Creation of mod-output/ and copy of final files into the game folder.

3- Game Files Modified by the Script

- data\ SOUND\CRI_DATA\ADX_SONG\ADX_SELECT.acb
- data\ SOUND\CRI_DATA\ADX_SONG\ADX_SELECT.awb
- data\ SOUND\CRI_DATA\ADX_SONG\AVEX.acb
- data\ SOUND\CRI_DATA\ADX_SONG\AVEX.awb
- data\flash\data_jp\common\bgm\bgmtex.pac
- data\flash\data_jp\menu\config\config_extex.pac
- data\flash\data_jp\menu\config\configtex.pac

4- Preparing Your Songs

Put your source music files in: new-musics/

Supported formats: WAV, FLAC, MP3, OGG.

Important: Files must be named with the in-game ID from 01 to 26. The script uses these IDs and the mapping table in this guide to write the correct HCA filenames in AVEX and ADX_SELECT.

```
01 artist - title.flac
02 artist - title.mp3
26 artist - title.ogg
```

5- Manual or Automatic WAV Preparation

When the script reaches the audio processing step, you have two options:

- Automatic path: press Enter. If tmp/new-wav is empty, the script generates WAV files from new-musics with ffmpeg.
- Manual path: before pressing Enter, normalize/loudness-fix the tracks in Audition, Audacity, or another editor, then export WAV files to tmp/new-wav.

Manual loudness recommendation: -6 LUFS / -0.1 dBTP. The script deletes Adobe Audition .pkf cache files from tmp/new-wav after the manual step.

6- Preparing Song Previews

- Optional preview files go in new-musics-previews/.
- Supported formats: WAV, FLAC, MP3, OGG.
- Use the same in-game IDs as the main songs: 01 to 26.
- If new-musics-previews is empty, the script creates 50-second preview WAVs from tmp/new-wav in tmp/new-wav-prev.
- Preview HCA files are encoded without looping.

7- Optional UI Texture Files

If you want custom song titles/artwork in the game UI, put any of these PAC files in new-musics-gfx/:

```
data\flash\data_jp\common\bgm\bgmtex.pac
data\flash\data_jp\menu\config\config_extex.pac
data\flash\data_jp\menu\config\configtex.pac
```

- bgmtex.pac is used for the music selection sprites.
- config_extex.pac and configtex.pac are used by the factory sprites.
- The script backs up the original files, copies custom files into the game folder when present, and includes them in mod-output/.

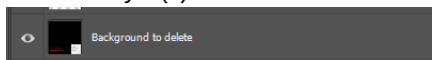
Font

Install .\tools\FOT-NewRodin Pro DB.otf in your windows system.

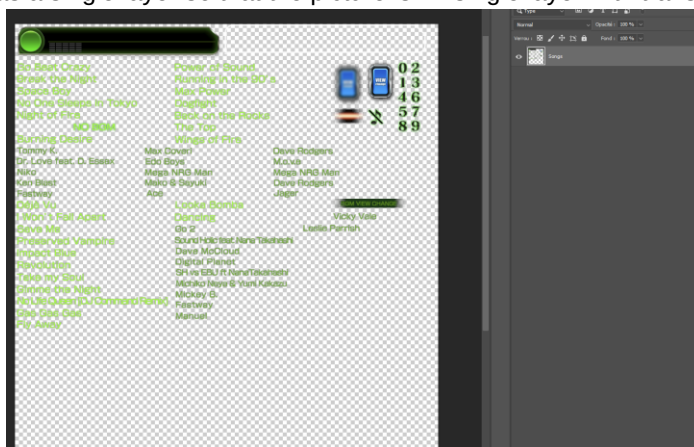
Use

Photoshop

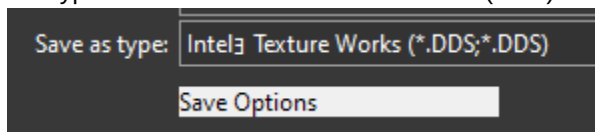
- You will find “ready to modify” templates in .\tools\
- Install “Intel Texture Works” Photoshop plugin for editing DDS
- Open the template you want to work on from .\tools\ and replace Artists and Titles with your songs ones
- At the end before exporting, delete the “to delete” layer(s):



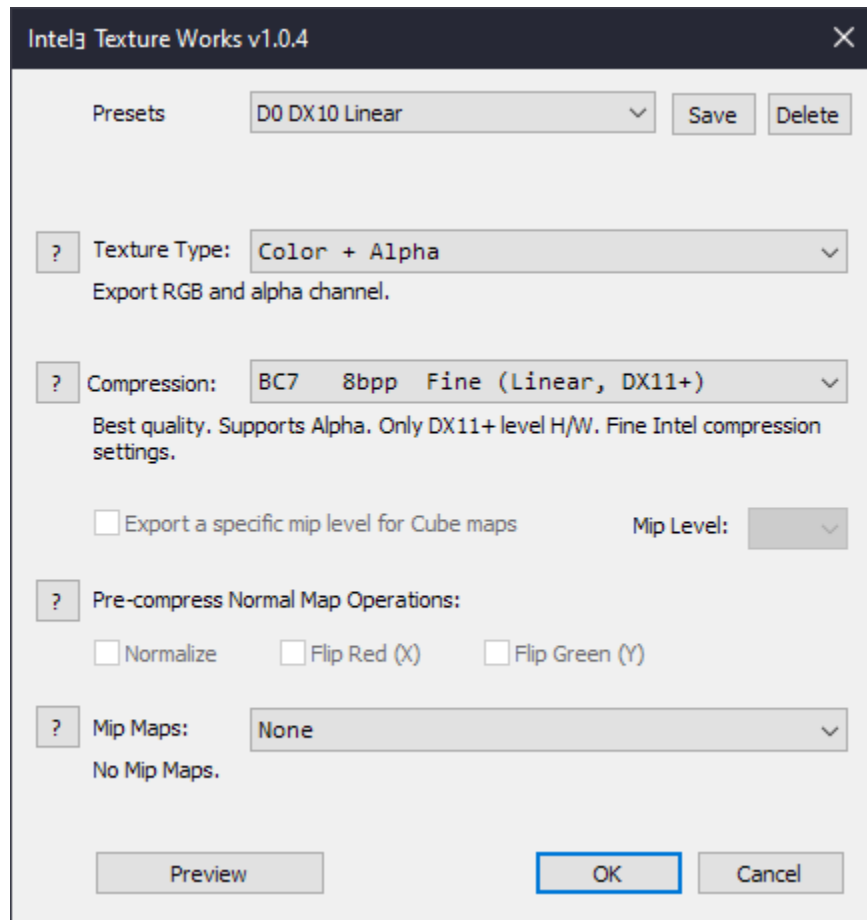
- Then merge all other layers as a single layer so that the picture is : 1 single layer with transparent background



- Then when Press Save As, in the Filetype Picker select Intel Texture Works (DDS) like so:



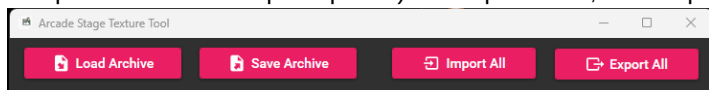
- It will show you the export options screen. Pick “Color + Alpha” for the Texture Type, “BC7 8bpp Fine (Linear, DX11+)” for the Compression, and “None” for Mip Maps, like in the following screenshot:
(any other formats will either result in the wrong file size, or in the case of sRGB lead to wrong colors ingame. Therefore, make sure to pick the correct type to avoid issues like that.)



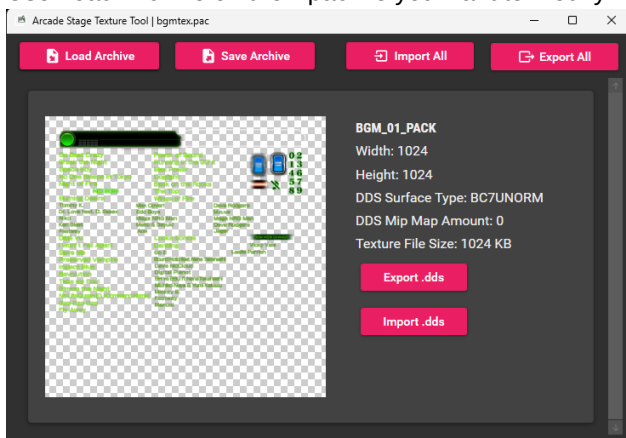
- Click okay to save your file.

ArcadeStageTextureTool

- You need to use that tool on the .\new-music-gfx*.pac files to extract the .dds files (not necessary if you use the provided Photoshop templates) or to open them, then import the new .dds files.



- Use Load Archive on the *.pac file you want to modify



- Then Import .dds to replace the sprite with the .dds file saved from Photoshop
- Then Save Archive to save the .pac file

Note: You can use the same .dds file for both: config_extex.pac and configtex.pac

8- Running the Script

1. Run `initial_d_arcade_stage_zero_v2_bgm_automod.cmd`.
2. Select the game directory. The script saves it for the next run.
3. Choose a backup option: full backup, missing files only, restore backup, or skip and continue.
4. Let the script extract `ADX_SELECT` and `AVEX` into `tmp/` folders.
5. At the audio step, either press Enter for automatic WAV generation or prepare normalized WAVs manually in `tmp/new-wav`.
6. Wait for full songs and previews to be converted to HCA.
7. Optionally prepare custom PAC artwork in `new-musics-gfx` when the script pauses for the graphics step.
8. Let the script rebuild `ACB/AWB`, publish files to the game folder, and create `mod-output/`.

9- Final Output

The final package is created in:

```
mod-output/
```

It contains the same folder structure as the game files:

```
mod-output/data/SOUND/CRI_DATA/ADX_SONG/ADX_SELECT.acb
mod-output/data/SOUND/CRI_DATA/ADX_SONG/ADX_SELECT.awb
mod-output/data/SOUND/CRI_DATA/ADX_SONG/AVEX.acb
mod-output/data/SOUND/CRI_DATA/ADX_SONG/AVEX.awb
mod-output/data/flash/data_jp/common/bgm/bgmtex.pac
mod-output/data/flash/data_jp/menu/config/config_extex.pac
mod-output/data/flash/data_jp/menu/config/configtex.pac
```

The script also copies the generated files over the selected game directory to apply the patch immediately.

10- Song ID Mapping

Use IDs 01-26 in new-musics and new-musics-previews. The script maps those IDs to the actual extracted AVEX and ADX_SELECT HCA filenames listed below.

Annex - Base Game Songs and HCA Mapping

ID	Artist	Title	Context	Status	AVEX HCA	ADX_SELECT Preview HCA
01	clutcho	strobe	irohazaka	unlocked	00000_streaming.hca	00007_streaming.hca
02	月光グリーン	inner fighter	momiji line	unlocked	00001_streaming.hca	00008_streaming.hca
03	backdraft smiths	curious (d mix)	usui	unlocked	00002_streaming.hca	00009_streaming.hca
04	backdraft smiths	candle flames	akina	unlocked	00003_streaming.hca	00010_streaming.hca
05	clutcho	avoid	lake akina	unlocked	00004_streaming.hca	00011_streaming.hca
06	the valves	no matter	akagi	unlocked	00005_streaming.hca	00012_streaming.hca
07	backdraft smiths	crank it up	tsukuba	unlocked	00006_streaming.hca	00013_streaming.hca
08	backdraft smiths	we'll start our race (d version)	hakone	unlocked	00007_streaming.hca	00014_streaming.hca
09	clutcho	liberation	happo	unlocked	00008_streaming.hca	00015_streaming.hca
10	the hug me	one by one	myogi	unlocked	00009_streaming.hca	00016_streaming.hca
11	backdraft smiths	the brave (d version)	nagao	unlocked	00010_streaming.hca	00017_streaming.hca
12	m.o.v.e	gamble rumble		locked	00011_streaming.hca	00018_streaming.hca
13	ken blast	the top		locked	00012_streaming.hca	00019_streaming.hca
14	marko polo	stop your self control		locked	00013_streaming.hca	00020_streaming.hca
15	hely	forever sad	akina snow	unlocked	00015_streaming.hca	00022_streaming.hca
16	takenobu&kunoichi@sega	burn inside		locked	00014_streaming.hca	00021_streaming.hca
17	stephy martini	mad desire	sadamine	unlocked	00016_streaming.hca	00023_streaming.hca
18	richard cottle	set me free babe		unlocked	00025_streaming.hca	00025_streaming.hca
19	sachi@sega	chase the ghost		unlocked	00024_streaming.hca	00024_streaming.hca
20	njk record	crazy hot	touhou collab part 1	locked	00017_streaming.hca	00000_streaming.hca
21	njk record	my destiny	touhou collab part 1	locked	00020_streaming.hca	00003_streaming.hca
22	sound holic vs eurobeat union feat nana takahashi	pandora zone		locked	00021_streaming.hca	00004_streaming.hca
23	sound holic vs eurobeat union feat nana takahashi	no life queen [dj command remix]	touhou collab part 2	locked	00022_streaming.hca	00005_streaming.hca
24	sound holic feat nana takahashi	preserved vampire	touhou collab part 2	locked	00023_streaming.hca	00006_streaming.hca
25	sound holic vs eurobeat union feat nana takahashi & 709sec	00 heaven	touhou collab part 3	locked	00019_streaming.hca	00002_streaming.hca
26	eurobeat union	get in the groove	touhou collab part 3	locked	00018_streaming.hca	00001_streaming.hca

Annex - Quick Checklist

- new-musics contains 01-26 audio files, or the subset you want to replace.
- new-musics-previews optionally contains 01-26 preview audio files; otherwise previews are generated from tmp/new-wav.
- new-musics-gfx optionally contains bgmtex.pac, config_extex.pac, and/or configtex.pac.
- backup/ contains the original game ACB/AWB and PAC files before publishing.
- mod-output/ can be copied manually if you do not want to rely on the automatic publish step.